

Ammar Khairi

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Education

The University Of Edinburgh

Edinburgh, UK

M.Sc. in Artificial Intelligence; GPA: 7.0/10

Sep 2022 – Sep 2023

- Thesis: Summer Internship at Amazon Development Centre Scotland working on Efficient Fine-Tuning of Large Language Models for Multilingual Code Generation.

University of Khartoum

Khartoum, Sudan

B.Sc. in Electrical and Electronics Engineering; GPA: 8.64/10

Sep 2014 – Jun 2020

- Thesis: Text to Image Synthesis from Text Semantics using Attentional Generative Adversarial Networks

Work Experience

Research Scholar

Jan 2025 – Current

Cohere (Cohere Labs)

London, UK

- Working under the supervision of Sara Hooker and Julia Kreutzer on novel test-time scaling methods to optimize multilingual and multitask LLM inference via parallel sampling.
- Developing adaptive sampling and search strategies to improve inference quality in multilingual open-ended tasks and synthetic data generation.

Research Assistant

Apr 2024 – Dec 2024

Department of Computer Science, University of Warwick

Coventry, UK

- Working in the TIA Lab under Dr. Fayyaz Minhas in the field of computational pathology.
- Researching the development of foundational models to forecast patient trajectories based on multi-modal data.

Research Assistant (Part-time)

Mar 2024 – Dec 2024

Department of Land Economy, University of Cambridge

Cambridge, UK

- Assist in advanced spatial analysis and applications of machine learning algorithms for urban health research on the H2020 project 'Emotional Cities'.
- Responsibilities include data collection, preprocessing, and contributing to ongoing research publications.

Teaching Assistant

Jan 2023 – Jul 2023

School of Informatics, The University of Edinburgh

Edinburgh, UK

- Served as a TA for the Computer Graphics course, leading lab sessions and tutorials in C++, OpenGL, and Blender.

Machine Learning Engineer

Jul 2021 – Sep 2022

AmunData Data-Driven Research & Advisory

Khartoum, Sudan

- Proposed the process of re-engineering the company's crop monitoring platform to an open source in-house solution, which reduced running costs by 90% and enabled greater optimization.
- Architected automated data pipelines to ingest raw Sentinel-2 satellite imagery, process vegetation insights like NDVI, and store processed data then extract actionable agricultural insights for data-driven decision making.

Full Stack Developer

Dec 2020 – Jun 2021

CODE Software

Khartoum, Sudan

- Managed migration of Sudatel's Direct Carrier Billing platform from Java to GoLang, enabling more scalable concurrent implementations which resulted in 10% increase in revenue for Anghami subscription service.
- Optimized Anghami's MySQL database serving over 700,000 users to limit storage growth by redesigning schema, optimizing queries and the effective use of indexing.

Projects

Storm Over the Nile: the Arabic Twitter Discussion on the Civil War in Sudan | [GitHub](#) | [Report](#)

- Consulted on NLP research project at Clingendael Institute of International Relations, analyzing polarization in Twitter discussions surrounding the Sudan conflict.
- Delivered a comprehensive toolkit, including a labeled dataset of polarized tweets with categories, vector embeddings, a classification model, and a Streamlit-based visualization app.

Data-Efficient Image Multi-Lingual Image Captioning | [GitHub](#)

- Developed a compact and efficient framework for multilingual image captioning in English, German and Arabic as part of my MSc coursework.
- Improved English caption Rouge scores by 18% over prior work while matching baseline metrics for German and Arabic using only 20% of baseline training data.
- Our approach utilizes CLIP, GPT-2 and transformer adapters to optimize caption quality while minimizing data and compute requirements.

Quantifying the Impacts of Violent Conflict on Local Agricultural Production | [GitHub](#) | [Report](#)

- Developed methodology leveraging Sentinel-2 satellite imagery and XGBoost classification to estimate cultivated agricultural lands in Sudan, achieving 91% accuracy in predicting crop types and coverage.
- Orchestrated large-scale data collection, preprocessing, and labeling of 100+GB of satellite imagery.
- Generated insights indicating a 40% expected drop in output to highlight food security risks

Publications

- **When Life Gives You Samples: The Benefits of Scaling up Inference Compute** | [Preprint](#)
To appear at *Empirical Methods in Natural Language Processing (EMNLP)*, 2025
- **Scaling Down Multilingual Language Models of Code** | [GitHub](#) | [Poster](#)
Conference on Neural Information Processing Systems (NeurIPS) - Black in AI Workshop 2023.
- **Multi-Class Image Classification using Kasami Orthogonal Codes.** | [GitHub](#) | [Extended Abstract](#)
Northern Lights Deep Learning Conference (NLDL) 2022.
- **SemGAN: Text to Image Synthesis from Semantics using Attentional GANs** | [Demo](#) | [Paper](#)
International Conference on Computer, Control, Electrical, and Electronics Engineering (ICCCEE), 2021.
- **Modeling, Visualization, and Analysis of African Innovation Performance.** | [Extended Abstract](#)
International Conference on Learning Representation (ICLR), 2020.

Additional Training

Philosophy & Computer Science Summer School

University of Bayreuth, Germany

Participant

Jul 2023

- Participated in an interdisciplinary summer school exploring topics at the intersection of philosophy and computer science, including AI ethics, explainability, fairness, and privacy.
- Wrote a short reflection piece during the program on: “The Ethics of AI Art” | [Article](#)